

Test Cases

Test Case 1: Insertion of Contacts

Action: Insert "AAAA" with phone "555-1234".

Insert "BBBB" with phone "555-2345"

Insert "CCCC" with phone "555-3456".

Expected output: AAAA should be inserted as the root node.

BBBB should be inserted to the right of "AAAA".

CCCC should be inserted to the right of "BBBB".

Actual output: All is inserted as per the expected output.

Test Case 2: Search for contacts

Action: Search for "AAAA".

Expected output: It should return "555-1234".

Actual output: Returns "555-1234".

Test Case 3: Search for Non-Existent Contact

Action: Search for "DDDD".

Expected output: It should return none.

Actual output: Returns None.

Test Case 4: In-order Traversal

Action: Call `inorder_traversal()`

Expected output: AAAA: 555-1234, BBBB: 555-2345, CCCC: 555-3456.

Actual output: AAAA: 555-1234, BBBB: 555-2345, CCCC: 555-3456.

Test Case 5: Delete Contact with No Children

Action: Delete "CCCC".

Expected output: "CCCC" should be removed from the tree.

Actual output: CCCC is removed from the tree.

Test Case 6: Delete Contact with One Child

Action: Delete "BBBB"

Expected output: "CCCC" should replace "BBBB" in the tree.

Actual output: "CCCC" replaces "BBBB" in the tree.

Test Case 7: Delete Root Node (Two Children)

Action: Delete "BBBB" (root node).

Expected output: "CCCC" should become the new root node.

Actual output: "CCCC" becomes the new root node.

Test Case 8: Verification of deletion

Action: Call inorder_traversal() after deleting CCCC

Expected output: AAAA: 555-1234, CCCC: 555-3456.

Actual output: AAAA: 555-1234, CCCC: 555-3456.